EEE407 Tutorial 5

1. Better windows for a building add \$3/ft^2 of window but saves \$0.55/ft^2 per year in reduced heating, cooling and lighting costs. With a discount rate of 12%:
    a. What is the net present value (NPV) of the better windows over a 30-year period with no escalation in the value of the annual savings?
    b. What is the internal rate of return (IRR) with no escalation rate?
    c. What is the NPV if the savings escalates at 7%/yr due to fuel savings?
    d. What is the IRR with that fuel escalation rate?

2. A 30-kW photovoltaic system on a building reduces the cumulative peak demand by 25 kW per year and reduces the annual electricity usage by 60,000 kWh/yr. The PV system costs \$35,000 to install, has no annual maintenance costs, and has an expected lifetime of 30 years. The utility rate charges \$0.07/kWh and \$9/kW demand charge.
    a. What annual savings in utility bills will the PVs deliver?
    b. What is the internal rate of return on the investment with no escalation in utility rates?
    c. If the annual savings on utility bills increases 6% per year, what is the IRR?

3. A small, 10-kW wind turbine that costs \$15,000 has a capacity factor of 30% and needs a fixed annual O&M cost of \$500 to keep the wind turbine operating in good condition. If it is paid for with a 7-%, 18-year loan, what is the LCOE of electricity generated (\$/kWh) by this system?

4. A 4-kW photovoltaic system that generates 8000 kWh/yr costs \$15,000. It is paid for with a 6%, 20-year loan.
    a. What is the capacity factor of this photovoltaic system?
    b. What is the cost of electricity from the PV system?
    c. With local utility electricity's levelized variable costs 11¢/kWh, at what rate would that price have to escalate over the 20-year period in order for the it to be the same as the cost of electricity from the PV system? Use Figure 5.4 and assume the buyer's discount rate is 15%.

5. A combustion turbine power plant has the following characteristics, find the LCOE.

    - Plant cost = $330 mil
    - Capacity = 350 MW
    - Capacity factor = 0.70
    - Heat rate = 9,905 Btu/kWh
    - Initial fuel cost = $3.90/mmBtu
    - Fixed O&M cost = $2.45 mil/yr
    - Variable O&M cost = $0.0045/kWh
    - Discount rate = 0.10/yr
    - Annual cost escalation rate = 0.06/yr
    - 25-year lifetime